

Machine Design II Project Report

Horizontal Wind Turbine Gearbox

REE 312

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Abstract

The gearbox in a horizontal-axis wind turbine is a critical mechanical component that connects the low-speed shaft from the rotor to the high-speed shaft driving the generator. Its primary function is to increase rotational speed from the turbine blades to the required input speed of the generator. This report presents the design, analysis, and verification of a multi-stage helical gearbox tailored for a 55 kW wind turbine operating under variable wind conditions. Key design considerations include gear geometry, material selection, shaft sizing, fatigue analysis, and bearing life estimation, all aiming to ensure mechanical efficiency, reliability, and extended service life. Analytical calculations were validated through CAD modeling and simulation using Autodesk Inventor. The resulting design meets performance and safety criteria, making it a viable solution for small- to medium-scale wind energy applications.

1 Introduction

As global energy demand continues to rise and the shift toward sustainable power sources becomes more urgent, wind energy has established itself as a key contributor in the renewable energy sector. Horizontal-axis wind turbines (HAWTs), in particular, are widely utilized due to their high efficiency and scalability. One of the most essential mechanical components in these turbines is the gearbox, which transmits power from the slowly rotating rotor shaft to the high-speed generator shaft.

This report presents the design and analysis of a multi-stage spur gear transmission system for a 55 kW horizontal wind turbine. The gearbox is responsible for stepping up the low rotational speed of the rotor to the much higher speed required by the generator, while ensuring reliable torque transmission and mechanical integrity.

The design process involves comprehensive gear force calculations, determination of shaft dimensions based on shear and bending stress, fatigue life assessment, and key sizing for secure torque transfer. Bearing selection was carried out based on radial load analysis and required bearing life using dynamic load ratings. Spur gears were chosen for their simplicity, ease of manufacturing, and efficiency in transmitting power over parallel shafts.

To validate the hand calculations and mechanical decisions, the design was modeled and analyzed using Autodesk Inventor. This integrated approach ensures a robust and manufacturable gearbox suitable for medium-scale wind turbine applications, contributing to reliable and clean energy generation.

2 Hand Calculations for Gear Forces and Torques

2.1 Gear Data and Speed Relations

Modules: Gear 2 and Gear 3 $\rightarrow m = 10$ mm; Gear 4 and Gear 5 $\rightarrow m = 6$ mm

$$D_2 = 10 \cdot 108 = 1080 \text{ mm}, \qquad n_2 = 95 \text{ rpm (CW)}$$
$$D_3 = 10 \cdot 24 = 240 \text{ mm}, \qquad n_3 = \frac{N_2}{N_3} \cdot n_2 = 427.5 \text{ rpm (CCW)}$$
$$D_4 = 6 \cdot 98 = 588 \text{ mm}, \qquad n_4 = 427.5 \text{ rpm (CCW)}$$
$$D_5 = 6 \cdot 28 = 168 \text{ mm}, \qquad n_5 = 1500 \text{ rpm (CW)}$$

Gear 2 Calculations

$$F_{32}^t = \frac{60,000 \cdot H}{\pi \cdot D_2 \cdot n_2} = \frac{60,000 \cdot 55}{\pi \cdot 1080 \cdot 95} = 10.238 \text{ kN}$$
$$F_{32}^r = F_{32}^t \cdot \tan(\phi) = 10.238 \cdot \tan(20^\circ) = 3.76263 \text{ kN}$$
$$T_2 = F_{32}^t \cdot \frac{D_2}{2} = 10.238 \cdot \frac{1080}{2} = 5528.54 \text{ N.m (CW)}$$

Gear 2 Reaction Forces

$$F_{a2x} = F_{32}^t = 10.238 \text{ kN}$$
$$F_{a2y} = F_{32}^r = 3.76263 \text{ kN}$$
$$T_3 = F_{32}^r = 3.76263 \text{ kN}$$

Gear 3 Calculations

From the transmitted power and speed:

Force Transmission to Gear 3:

$$F_{23}^t = F_{32}^t = 10.238 \text{ kN}$$
$$F_{23}^r = F_{32}^r = 3.76263 \text{ kN}$$

Bearing Reactions at Gear 3:

$$F_{b3x} = F_{23}^t = 10.238 \text{ kN}$$
$$F_{b3y} = F_{23}^r = 3.76263 \text{ kN}$$

Torque at Gear 3 (with corrected pitch diameter):

$$T_3 = F_{23}^t \cdot \left(\frac{D_3}{2}\right) = 10.238 \cdot \left(\frac{240}{2}\right) = 1228.56 \text{ kN.mm} = 1228.56 \text{ N.m (CW)}$$

Gear 4 Calculations

$$T_4 = T_3 = 1228.56 \text{ N.m (CCW)}$$
$$F_{54}^t = \frac{T_4}{\frac{D_4}{2}} = \frac{1228.56}{294} = 4.17878 \text{ kN}$$
$$F_{54}^r = F_{54}^t \cdot \tan(\phi) = 4.17878 \cdot \tan(20^\circ) = 1.52095 \text{ kN}$$

Gear 4 Reaction Forces

$$F_{b4x} = F_{54}^t = 4.17878 \text{ kN}$$
$$F_{b4y} = F_{54}^r = 1.52095 \text{ kN}$$

Gear 5 Calculations

$$F_{45}^t = F_{54}^t = 4.17878 \text{ kN}$$
$$F_{45}^r = F_{54}^r = 1.52095 \text{ kN}$$
$$T_5 = F_{45}^t \cdot \frac{D_5}{2} = 4.17878 \cdot \frac{168}{2} = 351.01752 \text{ N.m (CW)}$$

3 Factor of Safety Against Bending and Wear for Spur Gears

The AGMA design procedure is used to determine the safety factors for both bending fatigue and surface wear in spur gears. Below are the detailed calculations for both gear stages based on transmitted torque, geometry, material properties, and loading conditions.

3.1 General Assumptions and Data

- AGMA Quality Number: $Q_v = 6$
- Constants:

$$B = 0.25(12 - Q_v)^{2/3} = 0.8255, \quad A = 50 + 56(1 - B) = 59.773$$

3.2 Stage 1 Calculations (Gears 2 and 3)

Step 1: Pitch Line Velocity

$$V = \omega r = 95 \cdot \frac{540}{1000} \cdot \frac{\pi}{3} = 5.372 \text{ m/s}$$

Step 2: Dynamic Factor

$$K_v = \frac{59.773 + \sqrt{200 \cdot 5.372}}{59.773} = 1.435$$

Step 3: Geometry and Load Factors

$$d_p = \frac{1080}{25.4} = 42.51969 \text{ in}, \quad F = 4.724 \text{ in}, \quad b = 1 \text{ in}, \quad K_o = K_s = K_B = 1$$

Step 4: Dynamic Load Factor K_m

$$Cma = 0.127 + 0.0158(4.724) + (-0.93 \cdot 10^{-4})(4.724)^2 = 0.2033$$

$$Cpf = \frac{4.724}{10 \cdot 42.51969}(-0.0375 + 0.0125 \cdot 4.724) = 0.033$$

$$K_m = 1 + 1(Cpf \cdot 1.1 + Cma \cdot 1) = 1.2073$$

Step 5: Contact Stress

$$I = \frac{\cos(20)\sin(20)}{2} \cdot \left(\frac{0.222222}{0.2222222 + 1}\right) = 0.0292$$

$$\sigma_c = Z_E\sqrt{\frac{W_t \cdot K_o K_s K_m K_v}{d_p \cdot b \cdot I}} = 191 \cdot \sqrt{\frac{10238 \cdot 1.2073 \cdot 1.435}{42.51969 \cdot 1 \cdot 0.0292}} = 413.54 \text{ MPa}$$

Step 6: Bending Stress

$$\sigma_{b1} = \frac{W_t K_o K_s K_m K_v}{d_p \cdot b} \cdot \frac{1}{Y_j} = \frac{10238 \cdot 1.2073 \cdot 1.435}{120} \cdot \frac{1}{0.44} = 33.59 \text{ MPa}$$

$$\sigma_{b2} = \frac{\ldots}{\ldots} \cdot \frac{1}{0.35} = 42.231 \text{ MPa}$$

Step 7: Material and Life Factors

$$S_t = 360 \cdot 1.3 = 468 \text{ MPa}, \quad S_c = 1332.2 \text{ MPa}$$

$$Y_N1 = 1.003, \quad Y_N2 = 0.9768, \quad Z_N1 = 0.982, \quad Z_N2 = 0.948, \quad Z_W = 1$$

Step 8: Factors of Safety

$$SF_1 = \frac{S_t \cdot Y_N1}{Y_\theta \cdot Y_z \cdot \sigma_{b1}} = \frac{468 \cdot 1.003}{1 \cdot 0.85 \cdot 33.59} = 12.65$$

$$SF_2 = \frac{468 \cdot 0.9768}{1 \cdot 0.85 \cdot 42.231} = 9.8$$

$$SH_1 = \frac{1332.2 \cdot 0.982}{0.85 \cdot 413.54} = 3.7, \quad SH_2 = \frac{1332.2 \cdot 0.948}{0.85 \cdot 413.54} = 3.6$$

3.3 Stage 2 Calculations (Gears 4 and 5)

Step 1: Pitch Line Velocity

$$V = \omega r = 427.5 \cdot \frac{\pi}{3} \cdot \frac{294}{1000} = 13.162 \text{ m/s}$$

Step 2: Dynamic Factor

$$K_v = \frac{59.773 + \sqrt{200 \cdot 13.162}}{59.773} = 1.668$$

Step 3: Geometry and Load Parameters

$$d_p = \frac{588}{25.4} = 23.1496 \text{ in}, \quad F = 2.835 \text{ in}, \quad b = 1 \text{ in}$$

Step 4: Dynamic Load Factor K_m

$$Cma = 0.127 + 0.0158(2.835) + (-0.93 \cdot 10^{-4})(2.835)^2 = 0.171$$

$$Cpf = \frac{2.835}{10 \cdot 23.1496}(-0.0375 + 0.0125 \cdot 2.835) = 0.01$$

$$K_m = 1 + 1(0.01 \cdot 1.1 + 0.171 \cdot 1) = 1.182$$

Step 5: Contact Stress

$$I = \frac{\cos(20)\sin(20)}{2} \cdot \left(\frac{0.28571}{0.285714 + 1}\right) = 0.0357$$

$$\sigma_c = 191 \cdot \sqrt{\frac{4178.7 \cdot 1.182 \cdot 1.668}{23.1496 \cdot 1 \cdot 0.0357}} = 445.94 \text{ MPa}$$

Step 6: Bending Stress

$$\sigma_{b4} = \frac{4178.7 \cdot 1.182 \cdot 1.668}{72} \cdot \frac{1}{0.44} = 30.714 \text{ MPa}$$

$$\sigma_{b5} = \frac{4178.7 \cdot 1.182 \cdot 1.668}{72} \cdot \frac{1}{0.38} = 35.56 \text{ MPa}$$

Step 7: Material and Life Factors

$$S_t = 360 \cdot 1.3 = 468 \text{ MPa}, \quad S_c = 1332.2 \text{ MPa}$$

$$Y_N4 = 0.9768, \quad Y_N5 = 0.955, \quad Z_N4 = 0.948, \quad Z_N5 = 0.9215$$

Step 8: Safety Factors

$$SF_4 = \frac{468 \cdot 0.9768}{0.85 \cdot 30.714} = 13.47, \quad SF_5 = \frac{468 \cdot 0.955}{0.85 \cdot 35.56} = 11.4$$
$$SH_4 = \frac{1332.2 \cdot 0.948}{0.85 \cdot 445.94} = 3.33, \quad SH_5 = \frac{1332.2 \cdot 0.9215}{0.85 \cdot 445.94} = 3.24$$

4 Shaft Diameter Calculations

The shaft diameters were designed using fatigue analysis and modified Goodman criteria, considering alternating bending and torsional stresses.

4.1 Assumptions

- Material: AISI 1050, Hot Rolled
- $S_u = 1090$ MPa, $S_y = 793$ MPa
- Temperature $T = 425°C$, Surface finish: hot-rolled $a = 2.86,\ b = -0.65$
- Endurance limit estimate: $S'_e = 0.5S_u = 545$ MPa

4.2 Shaft 1

$$M = 697.045968 \text{ N.m}, \quad T = 5528.54 \text{ N.m}$$
$$K_a = 0.41, \quad d = 75 \text{ mm} \Rightarrow K_b = 0.766$$
$$K_c = 0.9, \quad K_d = K_e = 1.0$$
$$S_e = K_a K_b K_c S'_e = (0.41)(0.766)(0.9)(545) = 154.167 \text{ MPa}$$
$$q = 0.85, \quad \frac{r}{d} = 0.013 \Rightarrow \frac{D}{d} = 1.067 \Rightarrow K_t = 2.55, \quad K_{ts} = 1.75$$
$$K_f = 1 + q(K_t - 1) = 2.3175, \quad K_{fs} = 1 + q(K_{ts} - 1) = 1.801$$
$$A = \sqrt{4(K_f M)^2 + 3(K_{fs} T)^2} = \sqrt{4(2.3175 \cdot 697.045968)^2 + 3(1.801 \cdot 5528.54)^2} = 3230.80802$$
$$B = 3(1.801 \cdot 5528.54)^2 = 15967.50005$$

Using $n = 2.5$: $\quad 2.5 = \frac{\pi d^3}{16}\left(\frac{A}{S_e} + \frac{B}{S_u}\right)$

$$\Rightarrow d_1 = 76 \text{ mm}, \quad \textbf{Chosen:} \ \boxed{80 \text{ mm}}$$

4.3 Shaft 2

$$
\begin{aligned}
M &= 771.1335658 \text{ N.m}, \quad T = 1228.56 \text{ N.m} \\
d &= 70 - 75 \text{ mm} \Rightarrow K_b = 0.775 \\
K_a &= 0.41, \quad K_c = 0.9, \quad S'_e = 545 \Rightarrow S_e = 155.8564 \text{ MPa} \\
\frac{r}{d} &= 0.0143, \quad q = 0.85 \Rightarrow K_t = 2.55, \; K_{ts} = 1.75 \\
K_f &= 1 + 0.85(2.55 - 1) = 2.3175, \quad K_{fs} = 1 + 0.89(1.75 - 1) = 1.801 \\
A &= 4(K_f M)^2 = 3574.2041 \\
B &= 3(K_{fs} T)^2 = 3832.4 \\
\text{Using } n = 2.5: \quad 2.5 &= \frac{\pi d^3}{16}\left(\frac{A}{S_e} + \frac{B}{S_u}\right) \\
\Rightarrow d_2 &= 69.57 \text{ mm}, \quad \textbf{Chosen: } \boxed{75 \text{ mm}}
\end{aligned}
$$

4.4 Shaft 3

$$
\begin{aligned}
M &= 233.34133 \text{ N.m}, \quad T = 351.01752 \text{ N.m} \\
d &= 50 \text{ mm} \Rightarrow K_b = 0.816 \\
K_a &= 0.41, \quad K_c = 0.9, \quad S_e = 164.102 \text{ MPa} \\
K_t &= 2.4, \quad K_{ts} = 1.65, \quad q = 0.85, \quad q_s = 0.89 \\
K_f &= 1 + 0.85(2.4 - 1) = 2.19, \quad K_{fs} = 1 + 0.89(1.65 - 1) = 1.5785 \\
A &= 4(K_f M)^2 = 1022.035 \\
B &= 3(K_{fs} T)^2 = 959.697 \\
\text{Using } n = 2.5: \quad 2.5 &= \frac{\pi d^3}{16}\left(\frac{A}{S_e} + \frac{B}{S_u}\right) \\
\Rightarrow d_3 &= 44.9 \text{ mm}, \quad \textbf{Chosen: } \boxed{50 \text{ mm}}
\end{aligned}
$$

4.5 Key Design

The keys are designed to resist shear failure under transmitted torque using AISI 1030 Annealed steel:

$$S_y = 317 \text{ MPa}, \quad S_{ut} = 430 \text{ MPa}, \quad \tau_{allow} = 0.577 S_y = 182.909 \text{ MPa}$$

We assume two keys per connection: $n_f = 2$, and use:

$$F = \frac{T}{r}, \quad \tau = \frac{F}{n_f \cdot b \cdot L} \Rightarrow L = \frac{F}{n_f \cdot b \cdot \tau}$$

Key 2

$$T_2 = 5528.54 \text{ N.m} = 5528540 \text{ N.mm}, \quad r = 540 \text{ mm} \Rightarrow F = \frac{5528540}{540} = 10228.0$$
$$b = 10 \text{ mm}$$
$$L_2 = \frac{10228.05}{2 \cdot 10 \cdot 182.909} = \boxed{59 \text{ mm}}$$

Standard used: $F = 120$ mm

Key 3

$$T_3 = 1228.56 \text{ N.m} = 1228560 \text{ N.mm}, \quad r = 120 \text{ mm} \Rightarrow F = \frac{1228560}{120} = 10238.0$$
$$b = 8 \text{ mm}$$
$$L_3 = \frac{10238.0}{2 \cdot 8 \cdot 182.909} = \boxed{14 \text{ mm}}$$

Standard used: $F = 120$ mm

Key 4

$$T_4 = 1228.56 \text{ N.m}, \quad r = 42.5 \text{ mm} \Rightarrow F = \frac{1228560}{42.5} = 28896.7 \text{ N}$$
$$b = 10 \text{ mm}$$
$$L_4 = \frac{28896.7}{2 \cdot 10 \cdot 182.909} = \boxed{14 \text{ mm}}$$

Standard used: $F = 72$ mm

Key 5

$$T_5 = 351.02 \text{ N.m} = 351017.52 \text{ N.mm}, \quad r = 25 \text{ mm} \Rightarrow F = \frac{351017.52}{25} = 14040.7$$
$$b = 6 \text{ mm}$$
$$L_5 = \frac{14040.7}{2 \cdot 6 \cdot 182.909} = \boxed{6.39 \text{ mm}}$$

Standard used: $F = 72$ mm

5 Bearing Selection

5.1 Hand-Calculated Bearing Life

To ensure reliability, deep groove ball bearings were selected and analyzed using the standard bearing life equation:

$$C_{10} = a_f \cdot F_D \cdot \left(\frac{X_D}{X_0 + (\theta - X_0)(1 - R_D)} \right)^{\frac{1}{a}}$$

Where:

- $a_f = 1.2$ (application factor)
- F_D = Equivalent dynamic bearing load (kN)
- $X_D = \frac{L_D \cdot n_D \cdot 60}{10^6}$, where L_D is desired life (hrs) and n_D is shaft speed (rpm)
- $X_0 = 0.02$, $\theta = 4.439$, $b = 1.483$ (life equation constants)
- R_D = Reliability = 0.98 (for 98%)

Bearing B1 (Shaft 1 - Right side)

$$F_D = \sqrt{(7.376)^2 + (2.360)^2} = 7.745 \text{ kN}, \quad X_D = \frac{60 \cdot 95 \cdot 10000}{10^6} = 57$$

$$C_{10} = 1.2 \cdot 7.745 \left(\frac{57}{0.02 + 4.439(1 - 0.98)} \right)^{\frac{1}{1.483}} = \boxed{51.366 \text{ kN}} \quad \text{(Used: } d_1 = 80 \text{ mm)}$$

Bearing B2 (Shaft 1 - Left side)

$$F_D = \sqrt{(2.861)^2 + (0.916)^2} = 3.305 \text{ kN}, \quad X_D = 57$$

$$C_{10} = 1.2 \cdot 3.305 \left(\frac{57}{0.02 + 4.439(1 - 0.98)} \right)^{\frac{1}{1.483}} = \boxed{19.930 \text{ kN}} \quad \text{(Used: } d_1 = 80 \text{ mm)}$$

Bearing B3 (Shaft 2 - Right side)

$$F_D = \sqrt{(8.233)^2 + (2.373)^2} = 8.5815 \text{ kN}, \quad X_D = \frac{60 \cdot 427.5 \cdot 10000}{10^6} = 256.5$$

$$C_{10} = 1.2 \cdot 8.5815 \left(\frac{256.5}{0.02 + 4.439(1 - 0.98)} \right)^{\frac{1}{1.483}} = \boxed{93.83 \text{ kN}} \quad \text{(Used: } d_2 = 75 \text{ mm)}$$

Bearing B4 (Shaft 2 - Left side)

$$F_D = \sqrt{(6.186)^2 + (0.167)^2} = 6.188 \text{ kN}, \quad X_D = 256.5$$

$$C_{10} = 1.2 \cdot 6.188 \left(\frac{256.5}{0.02 + 4.439(1 - 0.98)} \right)^{\frac{1}{1.483}} = \boxed{67.71 \text{ kN}} \quad \text{(Used: } d_2 = 75 \text{ mm)}$$

Bearing B5 (Shaft 3 - Left side)

$$F_D = \sqrt{(0.911)^2 + (0.312)^2} = 0.967 \text{ kN}, \quad X_D = \frac{60 \cdot 1496.25 \cdot 10000}{10^6} = 897.75$$

$$C_{10} = 1.2 \cdot 0.967 \left(\frac{897.75}{0.02 + 4.439(1 - 0.98)} \right)^{\frac{1}{1.483}} = \boxed{15.16 \text{ kN}} \quad \text{(Used: } d_3 = 50 \text{ mm)}$$

Bearing B6 (Shaft 3 - Right side)

$$F_D = \sqrt{(3.322)^2 + (1.209)^2} = 3.535 \text{ kN}, \quad X_D = 897.75$$

$$C_{10} = 1.2 \cdot 3.535 \left(\frac{897.75}{0.02 + 4.439(1 - 0.98)} \right)^{\frac{1}{1.483}} = \boxed{58.79 \text{ kN}} \quad \text{(Used: } d_3 = 50 \text{ mm)}$$

5.2 Selected Bearings

Based on the calculated dynamic load ratings C_{10}, required shaft diameters, and standard catalog availability, the following deep groove ball bearings were selected:

Bearing	C_{10} Required (kN)	Shaft Diameter (mm)	Selected Bearing
B1	51.37	80	6216
B2	19.93	80	6216
B3	93.83	75	6315
B4	67.74	75	6315
B5	15.16	50	6310
B6	58.79	50	6310

Table 1: Selected Bearings with Matching C_{10} Ratings and Shaft Sizes

All selected bearings are deep groove type, suitable for combined radial and limited axial loads. Sealed or shielded variants (e.g., 2RS1 or 2Z) were selected to minimize lubrication maintenance and to suit the enclosed gearbox environment.

6 Oil Seals, and Retaining ring Selection

6.1 oil seals

Oil seals are crucial components used to retain lubrication and exclude contaminants in rotating shafts. For this gearbox design, three different oil seals were selected to suit different shaft sizes and sealing requirements. The selections are listed below along with their dimensions and corresponding seal types:

- **Seal 1:** $75 \times 100 \times 10$ mm **Type: CRW1R**
- **Seal 2:** $45 \times 100 \times 10$ mm **Type: HMS5 RG**

6.2 Retaining Rings Selection

Retaining rings are used to axially secure components such as bearings and gears on shafts. Their selection is based on the shaft diameter D_s and the corresponding groove diameter D_g required for secure retention. Below is a detailed selection for each shaft.

Shaft 1

- **Bearing (Position)**: $D_s = 80$ mm $\Rightarrow D_g = 76.5$ mm
- **Gear 2 (Position)**: $D_s = 85$ mm $\Rightarrow D_g = 81.5$ mm

Shaft 2

- **Bearing (Position)**: $D_s = 75$ mm $\Rightarrow D_g = 72$ mm
- **Gears 3 and 4 (Position)**: $D_s = 80$ mm $\Rightarrow D_g = 76.5$ mm

Shaft 3

- **Bearing (Position)**: $D_s = 50$ mm $\Rightarrow D_g = 47$ mm
- **Gear 5 (Position)**: $D_s = 55$ mm $\Rightarrow D_g = 52$ mm

Selected Retaining Ring Types and Quantities

The chosen retaining rings from the DSHR series are as follows:

- $4 \times$ `DSHR-086`
- $1 \times$ `DSHR-085`
- $2 \times$ `DSHR-075`
- $2 \times$ `DSHR-047`
- $1 \times$ `DSHR-052`

These snap rings are chosen to accommodate the dimensional fit and axial load conditions required for each shaft element.
The selected seals are compatible with the shaft diameters and housing tolerances of the gearbox. CRW1R is a commonly used radial shaft seal with a metal case and a sealing lip design suitable for high-speed applications. The HMS5 RG is used for slightly different shaft geometry or operating conditions where reinforced rubber (RG) is preferred for flexibility and sealing efficiency.

7 Detailed Drawing

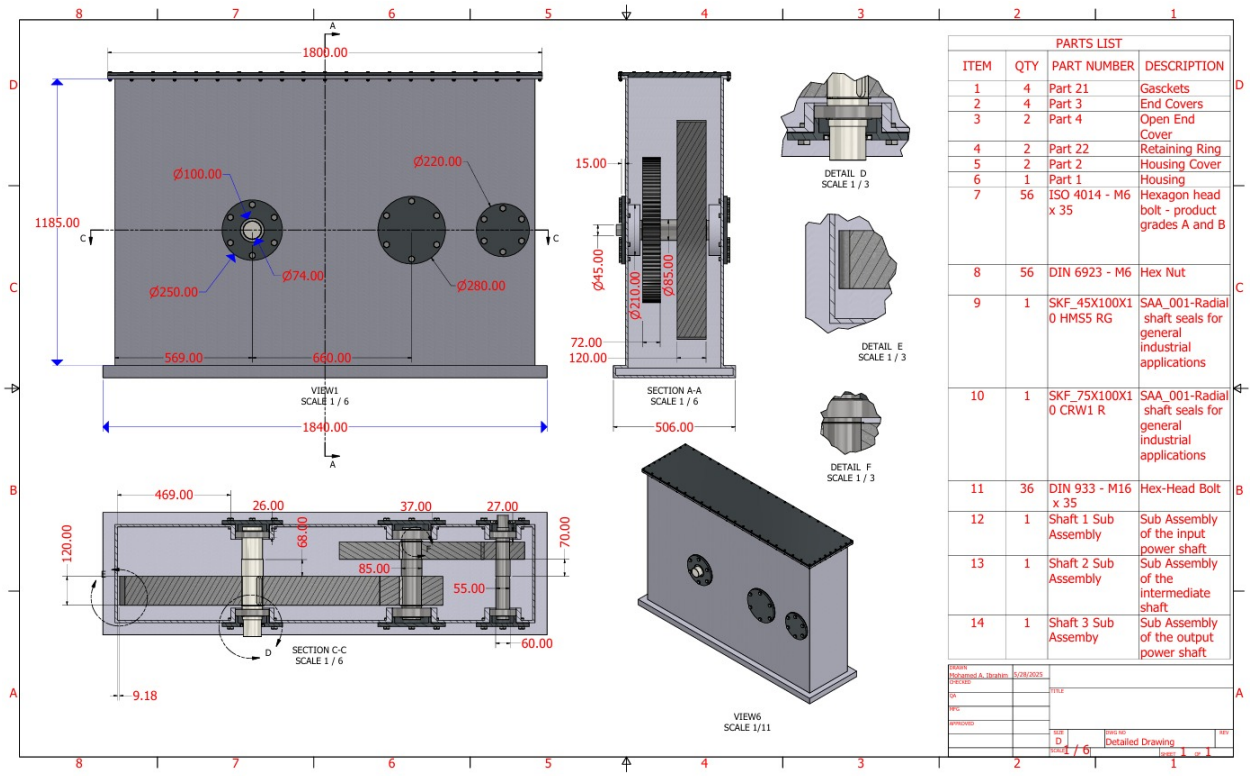


Figure 1: Gearbox Assembly Detailed Drawing and Bill of Materials

8 Conclusion

The design and analysis of the horizontal wind turbine gearbox have been successfully carried out in accordance with mechanical design standards and methodologies. The gearbox was developed using spur gears to transmit the required torque and speed through two stages, ensuring reliability, strength, and manufacturability.

Comprehensive calculations were performed to evaluate the factors of safety against both bending and wear for all gears involved. All gear components met or exceeded the desired safety margins, validating the structural integrity of the design. Additionally, shaft designs were optimized through fatigue analysis and verified by minimum diameter calculations, ensuring long-term durability under dynamic loading conditions.

Supporting systems such as oil seals, gaskets, and retaining rings were carefully selected to match the environmental and operational conditions of the gearbox, ensuring effective sealing and component retention.

Finally, all results were consolidated and validated through hand plots of shear force and bending moment diagrams for each shaft. A detailed drawing of the final assembly with a comprehensive bill of materials was included, reflecting a manufacturable and functional gearbox design.

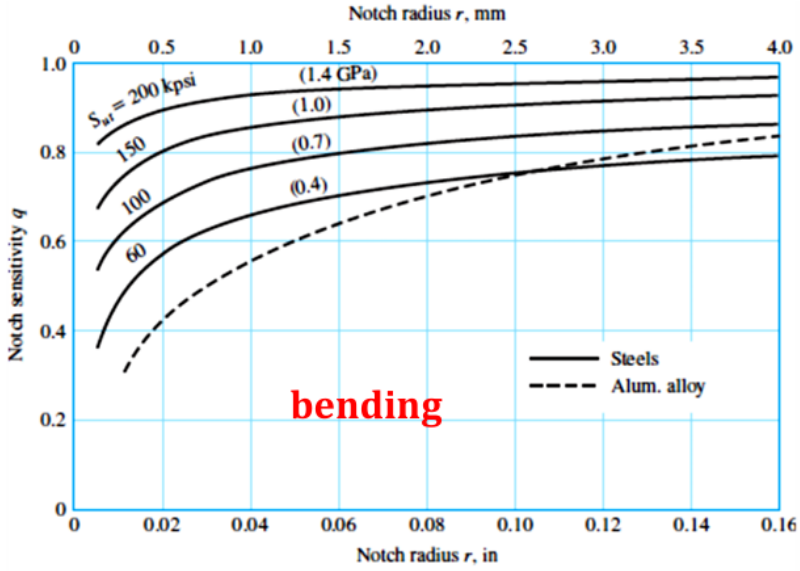
This report demonstrates a complete mechanical design workflow — from conceptualization and calculation to component selection and technical documentation — culminating in a robust gearbox system suitable for wind energy applications.

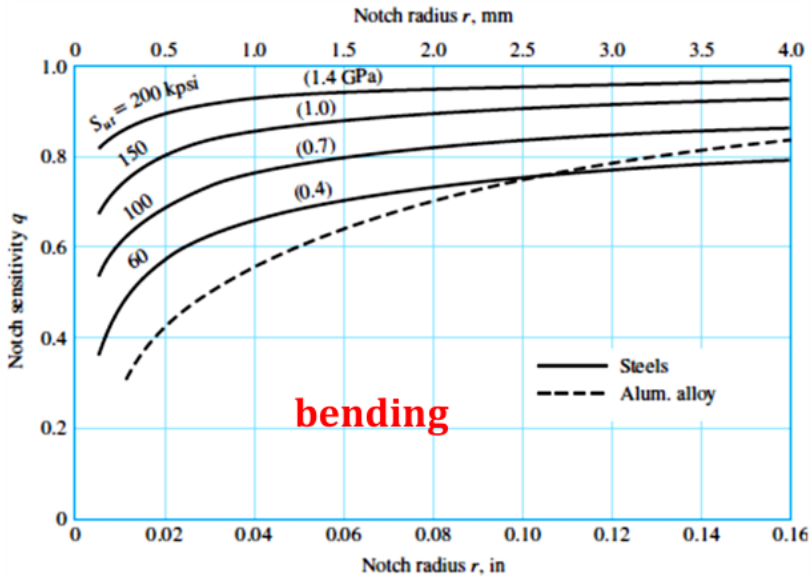
References

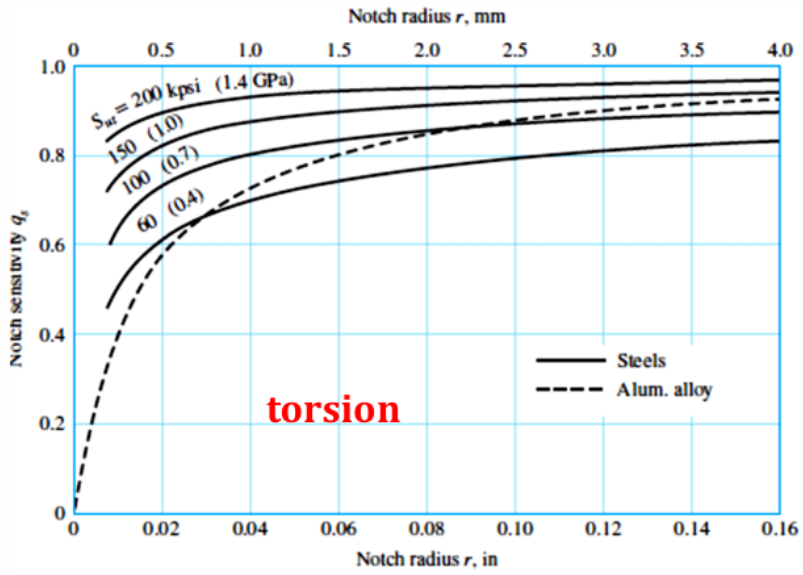
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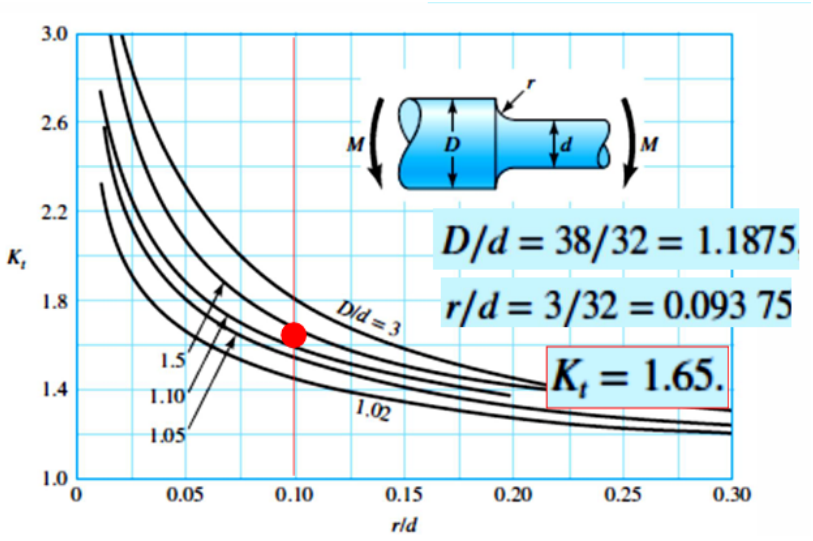
Appendices

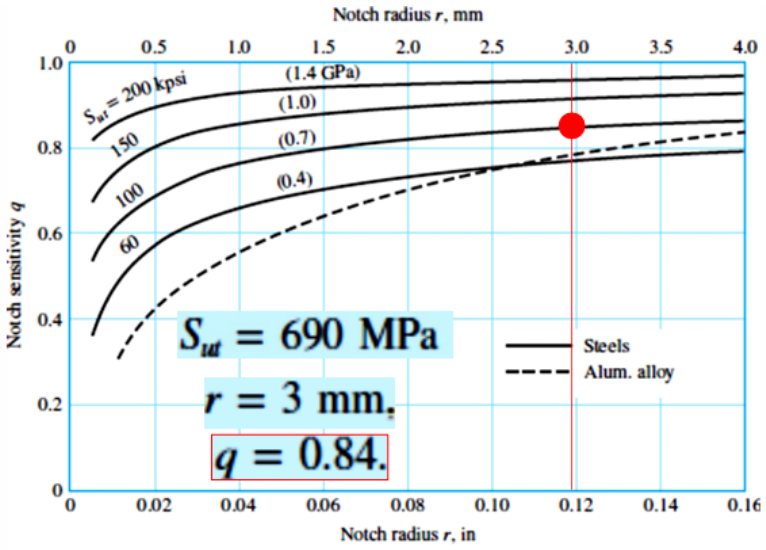
Surface Finish	Factor a		Exponent b
	S_{ut}, kpsi	S_{ut}, MPa	
Ground	1.21	1.38	−0.067
Machined or cold-drawn	2.00	3.04	−0.217
Hot-rolled	11.0	38.6	−0.650
As-forged	12.7	54.9	−0.758

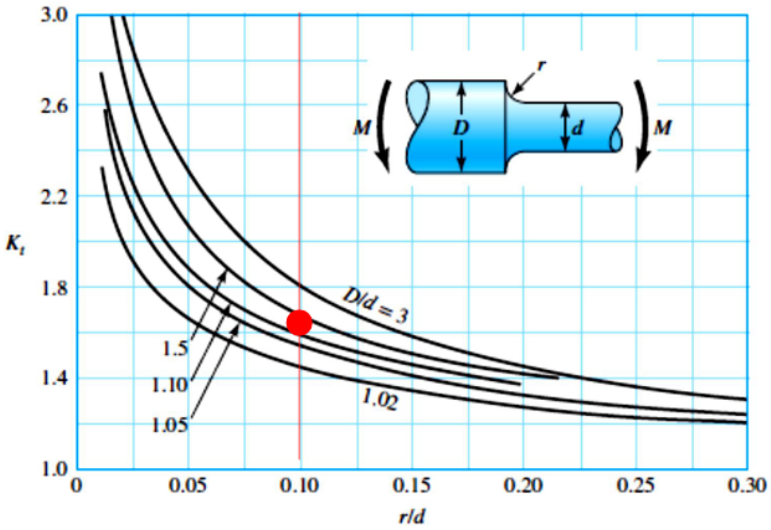


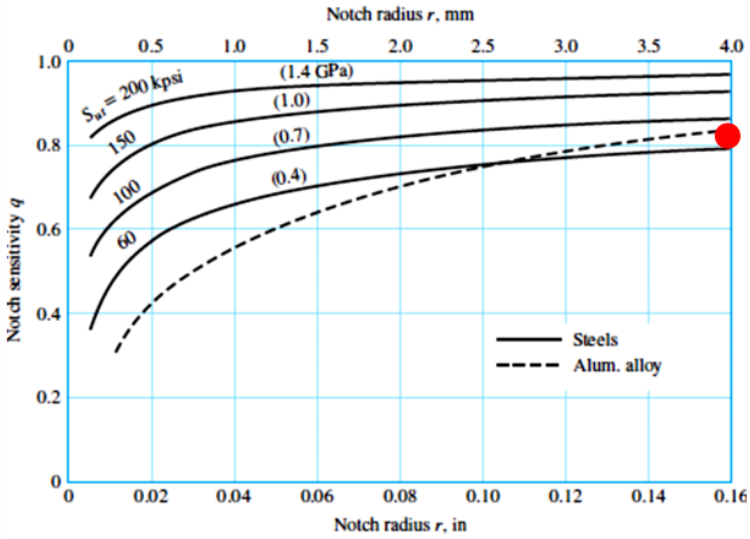


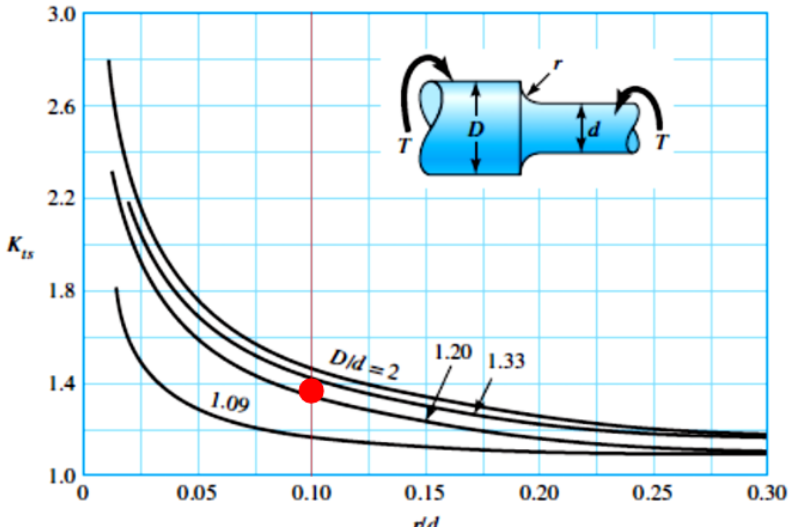

Notch radius r, mm
0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
1.0
0.8
0.6
0.4
0.2
0
Notch sensitivity q_s
S_{ut} = 200 kpsi (1.4 GPa)
150 (1.0)
100 (0.7)
60 (0.4)
Steels
Alum. alloy
torsion
0
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Notch radius r, in

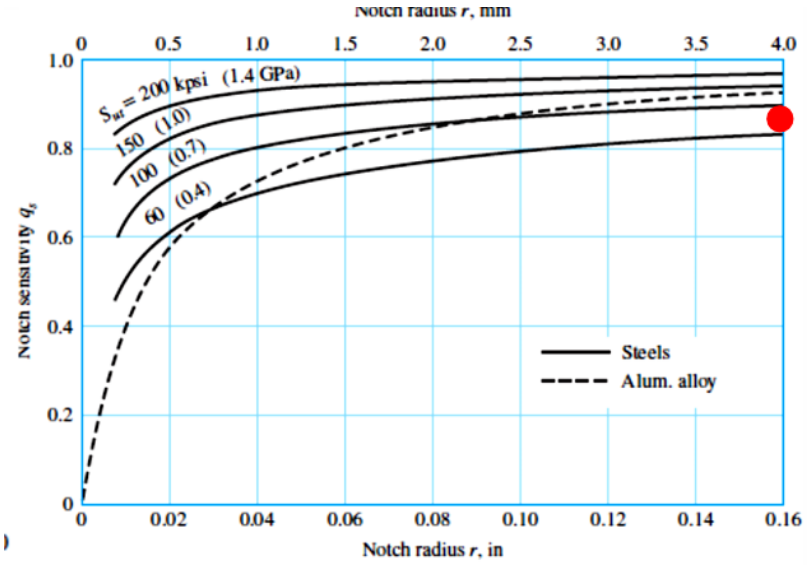

3.0
2.6
2.2
1.8
1.4
1.0
K_t
r
M
D
d
M
D/d = 3
1.5
1.10
1.05
1.02
0
0.05
0.10
0.15
0.20
0.25
0.30
r/d
$D/d = 38/32 = 1.1875$
$r/d = 3/32 = 0.093\,75$
$K_t = 1.65.$


Notch radius r, mm
0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
1.0
0.8
0.6
0.4
0.2
0
Notch sensitivity q
S_{ut} = 200 kpsi
150
100
60
(1.4 GPa)
(1.0)
(0.7)
(0.4)
S_{ut} = 690 MPa
r = 3 mm,
q = 0.84.
Steels
Alum. alloy
0
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Notch radius r, in


3.0
2.6
2.2
1.8
1.4
1.0
K_t
r
M
D
d
M
D/d = 3
1.5
1.10
1.05
1.02
0
0.05
0.10
0.15
0.20
0.25
0.30
r/d


Notch radius r, mm
0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
1.0
0.8
0.6
0.4
0.2
0
Notch sensitivity q
S_{ut} = 200 kpsi
150
100
60
(1.4 GPa)
(1.0)
(0.7)
(0.4)
Steels
Alum. alloy
0
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Notch radius r, in


3.0
2.6
2.2
1.8
1.4
1.0
K_{ts}
r
D
d
T
T
D/d = 2
1.20
1.33
1.09
0
0.05
0.10
0.15
0.20
0.25
0.30
r/d


Notch radius r, mm
0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
1.0
0.8
0.6
0.4
0.2
0
Notch sensitivity q_s
S_{ut} = 200 kpsi (1.4 GPa)
150 (1.0)
100 (0.7)
60 (0.4)
Steels
Alum. alloy
0 0.02 0.04 0.06 0.08 0.10 0.12 0.14 0.16
Notch radius r, in

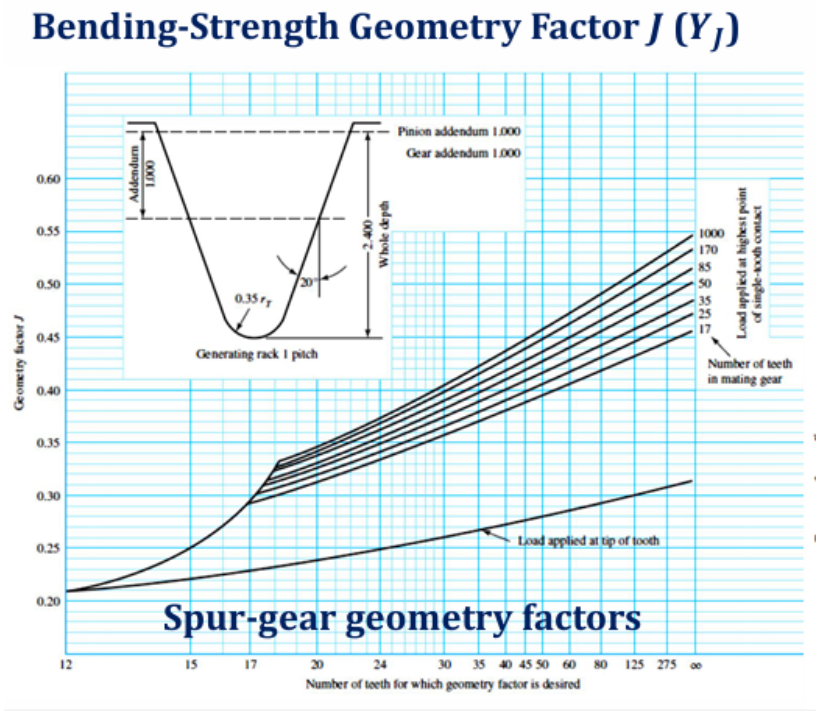

Bending-Strength Geometry Factor J (Y_J)
Pinion addendum 1.000
Gear addendum 1.000
Addendum 1.000
2.400 Whole depth
20°
0.35 r_T
Generating rack 1 pitch
Geometry factor J
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
1000
170
85
50
35
25
17
Number of teeth in mating gear
Load applied at highest point of single-tooth contact
Load applied at tip of tooth
Spur-gear geometry factors
12 15 17 20 24 30 35 40 45 50 60 80 125 275 ∞
Number of teeth for which geometry factor is desired

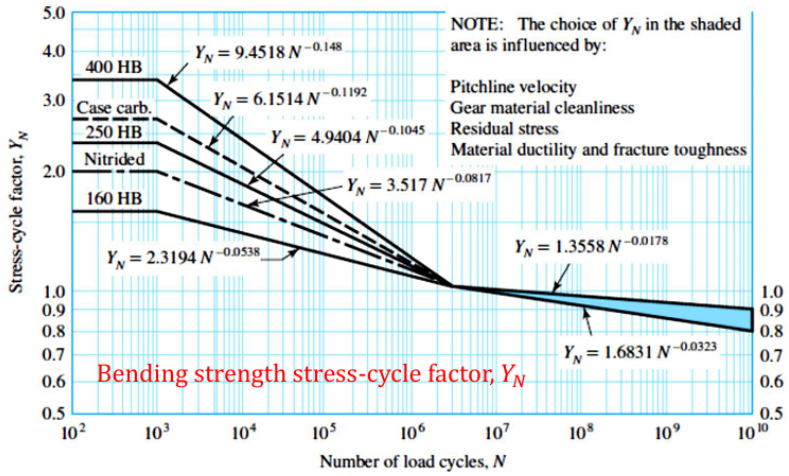
Table 14–8 **Elastic Coefficient C_p (Z_E), $\sqrt{\text{psi}}$ ($\sqrt{\text{MPa}}$)**

Pinion Material	Pinion Modulus of Elasticity E_p psi (MPa)*	Gear Material and Modulus of Elasticity E_G, lbf/in² (MPa)*					
		Steel 30×10^6 (2×10^5)	Malleable Iron 25×10^6 (1.7×10^5)	Nodular Iron 24×10^6 (1.7×10^5)	Cast Iron 22×10^6 (1.5×10^5)	Aluminum Bronze 17.5×10^6 (1.2×10^5)	Tin Bronze 16×10^6 (1.1×10^5)
Steel	30×10^6 (2×10^5)	2300 (191)	2180 (181)	2160 (179)	2100 (174)	1950 (162)	1900 (158)
Malleable iron	25×10^6 (1.7×10^5)	2180 (181)	2090 (174)	2070 (172)	2020 (168)	1900 (158)	1850 (154)
Nodular iron	24×10^6 (1.7×10^5)	2160 (179)	2070 (172)	2050 (170)	2000 (166)	1880 (156)	1830 (152)
Cast iron	22×10^6 (1.5×10^5)	2100 (174)	2020 (168)	2000 (166)	1960 (163)	1850 (154)	1800 (149)
Aluminum bronze	17.5×10^6 (1.2×10^5)	1950 (162)	1900 (158)	1880 (156)	1850 (154)	1750 (145)	1700 (141)
Tin bronze	16×10^6 (1.1×10^5)	1900 (158)	1850 (154)	1830 (152)	1800 (149)	1700 (141)	1650 (137)

Poisson's ratio = 0.30.

Pinion Material	Pinion Modulus of Elasticity E_p psi (MPa)*	Gear Material and Modulus of Elasticity E_G, lbf/in² (MPa)*					
		Steel 30×10^6 (2×10^5)	Malleable Iron 25×10^6 (1.7×10^5)	Nodular Iron 24×10^6 (1.7×10^5)	Cast Iron 22×10^6 (1.5×10^5)	Aluminum Bronze 17.5×10^6 (1.2×10^5)	Tin Bronze 16×10^6 (1.1×10^5)
Steel	30×10^6 (2×10^5)	2300 (191)	2180 (181)	2160 (179)	2100 (174)	1950 (162)	1900 (158)
Malleable iron	25×10^6 (1.7×10^5)	2180 (181)	2090 (174)	2070 (172)	2020 (168)	1900 (158)	1850 (154)
Nodular iron	24×10^6 (1.7×10^5)	2160 (179)	2070 (172)	2050 (170)	2000 (166)	1880 (156)	1830 (152)
Cast iron	22×10^6 (1.5×10^5)	2100 (174)	2020 (168)	2000 (166)	1960 (163)	1850 (154)	1800 (149)
Aluminum bronze	17.5×10^6 (1.2×10^5)	1950 (162)	1900 (158)	1880 (156)	1850 (154)	1750 (145)	1700 (141)
Tin bronze	16×10^6 (1.1×10^5)	1900 (158)	1850 (154)	1830 (152)	1800 (149)	1700 (141)	1650 (137)

Poisson's ratio = 0.30.



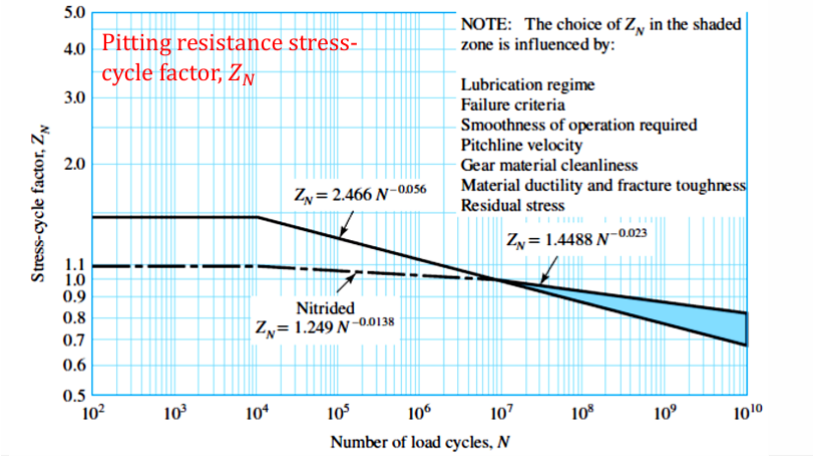


Table 14–10 Reliability Factors K_R (Y_Z)

Reliability	K_R (Y_Z)
0.9999	1.50
0.999	1.25
0.99	1.00
0.90	0.85
0.50	0.70

Condition	*A*	*B*	*C*
Open gearing	0.247	0.0167	$-0.765(10^{-4})$
Commercial, enclosed units	0.127	0.0158	$-0.930(10^{-4})$
Precision, enclosed units	0.0675	0.0128	$-0.926(10^{-4})$
Extraprecision enclosed gear units	0.00360	0.0102	$-0.822(10^{-4})$

1 AISI No.	2 Treatment	3 Temperature °C (°F)	4 Tensile Strength MPa (kpsi)	5 Yield Strength, MPa (kpsi)	6 Elongation, %	7 Reduction in Area, %	8 Brinell Hardness
1030	Q&T*	205 (400)	848 (123)	648 (94)	17	47	495
	Q&T*	315 (600)	800 (116)	621 (90)	19	53	401
	Q&T*	425 (800)	731 (106)	579 (84)	23	60	302
	Q&T*	540 (1000)	669 (97)	517 (75)	28	65	255
	Q&T*	650 (1200)	586 (85)	441 (64)	32	70	207
	Normalized	925 (1700)	521 (75)	345 (50)	32	61	149
	Annealed	870 (1600)	430 (62)	317 (46)	35	64	137
1040	Q&T	205 (400)	779 (113)	593 (86)	19	48	262
	Q&T	425 (800)	758 (110)	552 (80)	21	54	241
	Q&T	650 (1200)	634 (92)	434 (63)	29	65	192
	Normalized	900 (1650)	590 (86)	374 (54)	28	55	170
	Annealed	790 (1450)	519 (75)	353 (51)	30	57	149
1050	Q&T*	205 (400)	1120 (163)	807 (117)	9	27	514
	Q&T*	425 (800)	1090 (158)	793 (115)	13	36	444
	Q&T*	650 (1200)	717 (104)	538 (78)	28	65	235
	Normalized	900 (1650)	748 (108)	427 (62)	20	39	217
	Annealed	790 (1450)	636 (92)	365 (53)	24	40	187
1060	Q&T	425 (800)	1080 (156)	765 (111)	14	41	311
	Q&T	540 (1000)	965 (140)	669 (97)	17	45	277
	Q&T	650 (1200)	800 (116)	524 (76)	23	54	229
	Normalized	900 (1650)	776 (112)	421 (61)	18	37	229
	Annealed	790 (1450)	626 (91)	372 (54)	22	38	179
1095	Q&T	315 (600)	1260 (183)	813 (118)	10	30	375
	Q&T	425 (800)	1210 (176)	772 (112)	12	32	363
	Q&T	540 (1000)	1090 (158)	676 (98)	15	37	321
	Q&T	650 (1200)	896 (130)	552 (80)	21	47	269
	Normalized	900 (1650)	1010 (147)	500 (72)	9	13	293
	Annealed	790 (1450)	658 (95)	380 (55)	13	21	192
1141	Q&T	315 (600)	1460 (212)	1280 (186)	9	32	415
	Q&T	540 (1000)	896 (130)	765 (111)	18	57	262

1 AISI No.	2 Treatment	3 Temperature °C (°F)	4 Tensile Strength MPa (kpsi)	5 Yield Strength, MPa (kpsi)	6 Elongation, %	7 Reduction in Area, %	8 Brinell Hardness
4130	Q&T*	205 (400)	1630 (236)	1460 (212)	10	41	467
	Q&T*	315 (600)	1500 (217)	1380 (200)	11	43	435
	Q&T*	425 (800)	1280 (186)	1190 (173)	13	49	380
	Q&T*	540 (1000)	1030 (150)	910 (132)	17	57	315
	Q&T*	650 (1200)	814 (118)	703 (102)	22	64	245
	Normalized	870 (1600)	670 (97)	436 (63)	25	59	197
	Annealed	865 (1585)	560 (81)	361 (52)	28	56	156
4140	Q&T	205 (400)	1770 (257)	1640 (238)	8	38	510
	Q&T	315 (600)	1550 (225)	1430 (208)	9	43	445
	Q&T	425 (800)	1250 (181)	1140 (165)	13	49	370
	Q&T	540 (1000)	951 (138)	834 (121)	18	58	285
	Q&T	650 (1200)	758 (110)	655 (95)	22	63	230
	Normalized	870 (1600)	1020 (148)	655 (95)	18	47	302
	Annealed	815 (1500)	655 (95)	417 (61)	26	57	197
4340	Q&T	315 (600)	1720 (250)	1590 (230)	10	40	486
	Q&T	425 (800)	1470 (213)	1360 (198)	10	44	430
	Q&T	540 (1000)	1170 (170)	1080 (156)	13	51	360
	Q&T	650 (1200)	965 (140)	855 (124)	19	60	280